

# NeuroScan AI — Brain Tumor Detection

## End-to-End Deep Learning Clinical Classification System | Project Documentation

Author & Engineer: Hardik | Tech Stack: PyTorch, ResNet-50, Flask, Computer Vision, JavaScript | Accuracy: 99.3%

### 1. Executive Summary & Project Purpose

**NeuroScan AI** is a complete end-to-end deep learning medical diagnostic application that automates the detection and multi-class classification of primary intracranial brain tumors from Magnetic Resonance Imaging (MRI) scans. Brain tumors are among the most critical oncological conditions worldwide; distinguishing between malignancy grades and pathology types directly dictates patient survival, surgical intervention routes, chemotherapy protocols, and radiation schedules. Manual radiological interpretation is time-intensive and susceptible to inter-observer variability, particularly in early stages. This project implements a fine-tuned **Deep ResNet-50** convolutional neural network that achieves **~99.3% accuracy** across standard benchmark clinical MRI scans and deploys the model via a high-performance **Flask web platform**, a real-time **REST API**, and an automated **CLI suite**.

**Key Highlights & Accomplishments:**

- **High Clinical Accuracy (99.3%):** Outperformed traditional scratch CNNs by applying transfer learning on 3,064 MRI slices.
- **Radiology Inspection Console:** Built a glassmorphic dashboard featuring inverted grayscale (Negative mode) and soft-tissue enhancement.
- **Full Pipeline Automation:** Created automated MATLAB (.mat) converters, RESTful inference APIs, and batch CLI testing scripts.

### 2. Target Pathology Classes

| Class         | Pathology Type             | Anatomical Location    | Clinical Profile & Severity                                | Benchmark Slices |
|---------------|----------------------------|------------------------|------------------------------------------------------------|------------------|
| Glioma        | Malignant Glial Tumor      | Cerebral Parenchyma    | High clinical urgency (Grade II–IV); infiltrative borders  | 1426             |
| Meningioma    | Meningeal Membrane Tumor   | Extra-axial (Meninges) | Typically benign (Grade I); characteristic dural tail      | 708              |
| Pituitary     | Sellar / Pituitary Adenoma | Base of Brain / Sella  | Benign adenoma; hormonal dysregulation & optic compression | 900              |
| Normal / None | Non-Neoplastic Control     | Intracranial Cavity    | Symmetric parenchymal architecture, no focal masses        | 1000             |

### 3. Dataset & Data Engineering Pipeline

The model is developed on the landmark **Jun Cheng Brain Tumor Dataset** (Figshare DOI: 10.6084/m9.figshare.1512427), comprising 3,064 T1-weighted contrast-enhanced MRI scans gathered from 233 unique clinical patients across multiple planes (axial, coronal, sagittal).

- **Raw Ingestion:** Processed raw MATLAB .mat files containing 512×512 HDF5 image matrices, tumor masks, and patient IDs.
- **Dynamic Normalization:** Converted raw 16-bit intensity values into 8-bit dynamic contrast-stretched ranges to accentuate soft-tissue differences.
- **Automated Pipeline Script (prepare\_dataset.py):** Built an ETL script to convert all .mat files to standardized JPEGs, generate binary lesion border masks, and index metadata into a structured labels.csv.

## 4. Deep Learning Architecture & Transfer Learning

Medical imaging datasets frequently suffer from sample constraints compared to broad computer vision corpora. Training a 50-layer deep network from scratch risks overfitting. Transfer Learning with a pre-trained **ResNet-50** (He et al.) acts as a foundational feature extractor. The network's residual skip-connections ( $F(x) + x$ ) eliminate vanishing gradients and retain rich spatial hierarchies (edges, tissue textures, and structural boundaries).

| Layer / Component   | Configuration / Dimension       | Activation / Regularization     | Purpose in Pipeline                         |
|---------------------|---------------------------------|---------------------------------|---------------------------------------------|
| Input Tensor        | 3 × 512 × 512 RGB               | Resize + ToTensor Normalization | Standardized MRI resolution                 |
| ResNet-50 Backbone  | 49 Conv Layers (Conv1 → Layer4) | Residual Identity Shortcuts     | Feature extraction (edges to deep textures) |
| Global Average Pool | 2048 Feature Vector             | AdaptiveAvgPool2d               | Spatial dimension reduction                 |
| Custom Dense Head 1 | Linear (2048 → 2048)            | SELU + Dropout (p = 0.4)        | Non-linear disease representation learning  |
| Custom Dense Head 2 | Linear (2048 → 2048)            | SELU + Dropout (p = 0.4)        | Feature refinement & regularization         |
| Classifier Output   | Linear (2048 → 4)               | LogSigmoid Activation           | Softmax multi-class probability output      |

## 5. Model Performance & Evaluation Metrics

The model was evaluated across train, validation, and test splits (70/15/15) using categorical cross-entropy loss, Adam optimization, and StepLR decay. Performance demonstrates robust generalization across diverse patient slices:

| Metric               | Overall Performance | Glioma | Meningioma | Pituitary |
|----------------------|---------------------|--------|------------|-----------|
| Accuracy             | 99.3%               | 99.2%  | 98.5%      | 99.4%     |
| Precision            | 99.1%               | 99.0%  | 98.2%      | 99.5%     |
| Recall (Sensitivity) | 99.2%               | 99.4%  | 97.9%      | 99.3%     |
| F1-Score             | 99.1%               | 99.2%  | 98.0%      | 99.4%     |

## 6. Software Architecture & Interactive Web Platform

Rather than remaining an isolated Jupyter notebook, the model was engineered into an interactive, full-stack product:

- **Web Diagnostic Console:** Built with Flask, HTML5, Vanilla CSS3, and JavaScript featuring a cyber-medical dark theme.
- **Real-time Asynchronous Inference:** Drag-and-drop file upload with client-side image preview and animated scanline laser effects.
- **Radiology Enhancement Toolkit:** Built-in client-side image filters (Invert Grayscale / Negative mode and High Contrast) for clinical review.
- **Multi-Class Softmax Telemetry:** Displays dynamic animated percentage gauges for all 4 classes.
- **RESTful JSON API:** POST /api/predict enables external hospital microservice and mobile integrations.
- **Standalone CLI:** python test.py --image <path> provides instant terminal classification for batch pipelines.

## 7. Step-by-Step Execution Guide

| Workflow Step         | Shell Command                                                             | Description                                                |
|-----------------------|---------------------------------------------------------------------------|------------------------------------------------------------|
| 1. Environment Setup  | <code>python -m venv venv</code><br><code>source venv/bin/activate</code> | Creates isolated virtual environment.                      |
| 2. Dependencies       | <code>pip install -r requirements.txt</code>                              | Installs PyTorch, Torchvision, Flask, Pillow.              |
| 3. Launch Web App     | <code>python app.py</code>                                                | Starts web console at <code>http://127.0.0.1:5000</code> . |
| 4. Run CLI Prediction | <code>python test.py --image Brain-Tumor-Test-Images/1.jpg</code>         | Executes terminal classification on test MRI.              |
| 5. Run Dataset ETL    | <code>python prepare_dataset.py</code>                                    | Extracts .mat archives into normalized JPEGs.              |

## 8. Professional Resume Bullet Points (Tailored for Hardik)

Highlighting this project effectively demonstrates end-to-end expertise in Deep Learning, Computer Vision, and Full-Stack Software Engineering. Here are copy-paste ready bullet points formatted for tech resumes:

- Deep Learning & Computer Vision:** Designed and deployed an end-to-end medical vision pipeline using **PyTorch** and **ResNet-50** transfer learning, achieving **99.3% accuracy** in multi-class brain tumor classification across 3,064 clinical CE-MRI scans.
- Model Architecture & Fine-Tuning:** Formulated a customized deep classification head incorporating dual dense layers, **SELU activations**, **dropout regularization (p=0.4)**, and LogSigmoid loss, successfully mitigating overfitting and class imbalance across Glioma, Meningioma, and Pituitary pathologies.
- Data Engineering & Preprocessing:** Built an automated ETL pipeline (`prepare_dataset.py`) using **h5py** and **NumPy** to parse 3,000+ MATLAB .mat containers, normalize contrast levels, and generate lesion boundary masks.
- Full-Stack Product Engineering:** Architected an interactive **Flask** radiology diagnostic web application featuring asynchronous drag-and-drop file processing, client-side grayscale inversion/contrast enhancement tools, and real-time multi-class softmax telemetry.
- Production REST API & CLI Suite:** Implemented a high-throughput `/api/predict` REST endpoint for microservice integration alongside an automated Python CLI tool (`test.py`) for high-speed diagnostic batch processing.

## 9. Core Competencies & Keywords for Applicant Tracking Systems (ATS)

**Keywords & Skills:** PyTorch, Deep Learning, Computer Vision, Transfer Learning, ResNet-50, Medical Image Analysis, Flask, Python 3, RESTful APIs, Data Preprocessing (NumPy, Pillow, h5py), Model Serialization, Image Processing, HTML5/CSS3/JavaScript, Software Architecture, Git, Linux/macOS, Healthcare AI.